

综合目录使用方法

How to use catalog

跨页 Spread page

系列名称
Name of Series

FX6m
高硬度钢
FX6m Hardened Steels

系列特点
Feature of Series

高硬度材料加工用铣刀
Hardened Steels For Machining Endmills

用○、○标记可加工的材料
Machinable indication by ○

第一推荐
Strongly recommended

第二推荐
Recommended

型号 Model	规格 Size	涂层 Coating	刀数 Number of Flute	螺旋角 Helix Angle	精加工分类 Types of Finishing	加工用途 Applications	可对应加工材料 Work Material	规格数量 Number of Sizes	侧面 Side
高硬度钢用4刃平底铣刀 Hardened Steels 4-Flute Square Endmills									
φ1~φ12 X6m	45	无涂层	4	45°	精加工 Finishing	粗加工 Roughing	高硬度钢 Hardened Steels	31	○
在高硬度领域(48~65HRC)的长时间加工中也可实现稳定的长寿命。改进了刀刃形状,改善了倾斜量和加工精度。 Realized stably long tool life against high hardened steels (48 ~ 65HRC). Improved milling deflection and accuracy by advanced flute design.									
高硬度钢用2刃球头铣刀 Hardened Steels 2-Flute Ball Endmills									
R0.5~R6 X6m	30	无涂层	2	30°	精加工 Finishing	粗加工 Roughing	高硬度钢 Hardened Steels	27	○
高硬度材料用球头铣刀。最适用于精加工。适用于~65HRC的高硬度材料。 Ball end milling cutter for high hardness materials. Most suitable for finishing. Suitable for ~65HRC high hardness materials.									
高硬度钢用4刃圆鼻铣刀 Hardened Steels 4-Flute Corner Radius Endmills									
φ1xR0.05~φ12xR1 X6m	35	无涂层	4	35°	精加工 Finishing	粗加工 Roughing	高硬度钢 Hardened Steels	70	○
在高硬度领域(48~65HRC)的长时间加工中也可实现稳定的长寿命。增大了螺旋角,使得刀刃变得更锋利,而且圆鼻形状可以抑制崩刃。 Realized stably long tool life against high hardened steels (48 ~ 65HRC). Corner radius to prevent flute chipping, and high helix angle for cutting sharpness.									

刀刃形状
Shape of cutting edge

刀具外观图片
Tmodel Tool image

刀具特点
Feature of tool

粗加工
Roughing

半精加工
Semi-Finishing

精加工
Finishing

平面
Planing

侧面
Side Cutting

槽
Slotting

刻模
Die-sinking

曲面
Profiling

R加工
Radius

螺旋
Helical

产品页 Product page

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Machinable indication by ○

型号
Model

产品名称
Product name

刀具特点
Tool drawing

标准图
Feature of tool

FX6m
高硬度钢
FX6m Hardened Steels

高硬度钢用4刃平底铣刀
Hardened Steels 4-Flute Square Endmills

4刃·平底铣刀
4-Flute·Square

高硬度材料加工用。优异的切削排出性能,可实现高效率加工。
For machining of high-hardness materials. Excellent chip removal performance enables high efficiency.

1SDS3: 0~0.01
4SDS12: 0~0.015

0~0.005 (mm)

TYPE A:

TYPE B:

在高硬度领域(48~65HRC)的长时间加工中也可实现稳定的长寿命。
改进了刀刃形状,改善了倾斜量和加工精度。
Realized stably long tool life against high hardened steels (48 ~ 65HRC).
Improved milling deflection and accuracy by advanced flute design.

产品代码 Code No.	规格型号 Spec. Type	外径(D) Dia.	刀长(L) Length of Cut	螺旋角(H) Helix Angle	柄径(d) Shank Dia.	全长(L) Overall Length	形状 Type	定价 Retail Price
X6ME0104	1x3L	1	3	12°	4	50	A	
X6ME0104L	1x4L	1	4	12°	4	50	A	
X6ME0154	1.5x4.5L	1.5	4.5	12°	4	50	A	
X6ME0204	2x6L	2	6	12°	4	50	A	
X6ME02504	2.5x7L	2.5	7	12°	4	50	A	
X6ME0304	3x8L	3	8	12°	4	50	A	
X6ME0304	3x8L	3	8	12°	4	75	A	
X6ME0304D3	3x8L	3	8	12°	3	50	B	
X6ME0304D6	3x8L	3	8	12°	6	50	A	
X6ME0404	4x11	4	11	12°	4	50	B	

材料
Material

涂层
Coating

螺旋角
Helix Angle

刀数
Number of Flute

规格尺寸
Standard size

刀刃形状
Shape of cutting edge

平底
Square

球头
Ball

圆鼻
Round

切削参数参考表页

Recommended milling condition page

跨页 Spread page

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①可对应加工材料
Select work material

②可对应加工材料
Select tool size

切深量
Depth of Cut

(D: 刃径 Dia.)

③仅供参考的
切削参数
It is a reference of
cutting conditions

④请确认备注内容
Please check notes

※请使用刚性高、精度高的机床和夹具。
※加工参数会因切深量和机床刚性的状况而有所不同。请每次调整后在使用。
※请以相同的比率调整主轴转速和进给速度。
※建议使用吹气或油雾冷却方式。
※Use A Rigid And Precise Machine And Chuck Holder.
※Adjust milling conditions according to the volume of Depth of Cut and rigidity of the machine.
※Adjust both spindle speed and feed at the same rate.
※Use oilmist coolant or air blow.

备注栏的共同内容 Common notes for all products

说明 Explanation	本切削参数仅供参考。请根据实际的加工形状和所使用的机床等调整切削参数。 Recommended cutting conditions indicate just for reference. It should be adjusted according to milling shape and machine tool.
事前准备 Preparation	如果担心加工材料和铣刀会干涉时,请务必实际测量进行确认。 Tool measurement required. In order to avoid interference to the work material. 建议加工前充分考虑主轴的伸缩量和机床的特性。 Recommend to assess the machine characters, such as expansion of the spindle and others before using the tool. 建议采用顺铣加工方式。 Down-cut is recommended. 请尽量抑制刀具的偏摆量。(可能的话,请确认所用主轴转速下的动态偏摆精度。) Minimize chattering runout. (Recommend to measure actual runout at activated spindle speed.) 请使用刚性较大的铣刀刀柄和机床。 Use a rigid machine and holder. 拆装刀具或者测量刀具长度时请务必小心。 Extra care of handling when tool setting and measuring. 请根据需要控制刀具的伸出量在最短状态(不干涉的前提下)。 Minimize a possible tool overhang length.
调整参数 Milling condition adjustment	请在考虑加工材料及加工形状等的基础上,选用合适的切削油。 Select a cutting fluid appropriate to work material, milling shape and machining content. 加工R角等负载大的部位或复杂的形状时,请特别注意参数设定和刀路轨迹等。 When cutting high load sections or complex shapes, it requires attention to condition setting and tool path. 如果刀具发生崩刃,或者需要高精度加工时,请降低进给速度。 Reduce the feed for high accurate machining to avoid breakage of work piece. 发生振动或加工声音较吵等问题时,请以相同的比率降低主轴转速和进给速度。此外,机床主轴转速不足时,也请以相同的比率降低。 Reduce both spindle speed and feed at same rate for chattering, and also for insufficient spindle speed of a machine. 请根据机床刚性和工件的夹持状态等调整切削参数。实际加工时请根据加工形状、目的以及所用的机床等调整切削参数。 Adjust milling condition conforming with machine rigidity and clamping condition. Final milling Conditions are subject to machining profile, purpose and machine status. 主轴转速和进给速度会根据刀具伸出量和切深量的不同,会有大幅度变更的可能。 Spindle speed and feed are changed according to overhang length and depth of cut. 精加工时,请根据要求精度等调整切削参数。 For finishing process, please adjust to the optimized condition to meet the requirement of machining accuracy. 加工过程中切屑卡入、缠绕可能会有损加工面品质,因此请注意排屑。 Care for chip removal to avoid being stuck or caught during process for better surface quality.